
Algorithm 1: Estimating RDD with Cross-validated Bandwidth

Input: Dataset $\mathcal{D} = \{(Y_i, AdRank_i, FirstPosition_i)\}_{i=1}^n$ and the

train set $\mathcal{D}_{train}$ and test set $\mathcal{D}_{test}$ (50/50 split), cutoff c ,

candidate multipliers $\{0.25, 0.50, 1.00, 2.00\}$, and estimating

equation:

$$Revenues_i = \alpha + \beta FP_i + \gamma AdRank_i + \delta(FP_i \times AdRank_i) + \epsilon_i.$$

Output: Optimal bandwidth h_{optimal} , estimated treatment effect $\hat{\beta}$

1 Compute SD_{AdRank} = standard deviation of $AdRank_i$;

2 Set candidate bandwidths $\mathcal{H} = \{0.25, 0.50, 1.00, 2.00\} \times SD_{AdRank}$;

3 **foreach** $h \in \mathcal{H}$ **do**

4 Trim $\mathcal{D}_{train}$ to keep observations with $|AdRank_i| \leq h$;

5 Estimate the regression model on trimmed training data with the
estimating equation.

6 Trim $\mathcal{D}_{test}$ to keep observations with $|AdRank_i| \leq h$;

7 Predict $\hat{Y}_i$ for test set using estimated model;

8 Compute $MSE(h) = \frac{1}{n_{test}} \sum_{i \in \mathcal{D}_{test}} (\hat{Y}_i - Y_i)^2$;

9 Select $h_{\text{optimal}} = \arg \min_{h \in \mathcal{H}} MSE(h)$;

10 Trim full dataset to observations with $|AdRank_i| \leq h_{\text{optimal}}$;

11 Estimate the final regression model on trimmed data with the
estimating equation.

12 Return h_{optimal} and estimated $\hat{\beta}$;
